Year 11 Mathematics

Shape and Space 6

10.3 Quadrilaterals

Quadrilaterals

STARTER · 5 MIN

Date: _____ **Title:** Quadrilaterals

LEARNING OBJECTIVES

- Know the properties of special quadrilaterals
- Solve angle problems in quadrilaterals using algebra
- Prove properties of quadrilaterals

STARTER *Answer in your book. No calculators.*

1. Angles in a quadrilateral add up to?
2. Find the fourth angle: 90°, 100°, 80°
3. Solve $4x = 360$
4. How many lines of symmetry does a rectangle have?

Starter — Answers

STARTER

Tick or correct in green pen

1. **360°**

2. $360 - 270 = \mathbf{90°}$

3. $x = \mathbf{90}$

4. **2**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I know the properties of special quadrilaterals	☐
SILVER	I can solve angle problems in quadrilaterals using algebra	☐
GOLD	I can prove properties of quadrilaterals	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Quadrilaterals

INTRODUCTION · 3 MIN

The angles in any quadrilateral add up to 360°. Each special quadrilateral has its own properties.

Parallelogram

Opposite sides and angles are equal.

Diagonals bisect each other

Rhombus

Four equal sides.

Diagonals cross at 90°

Kite

Two pairs of equal adjacent sides.

One pair of equal angles

Trapezium

One pair of parallel sides.

Co-interior angles add to 180°

Examples 1 and 2 — Properties and Angles

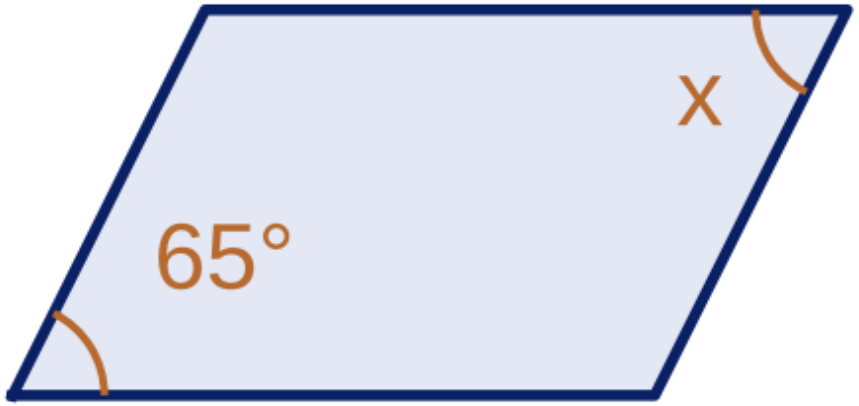
EXAMPLES · 5 MIN

BRONZE *Watch and copy into your book*

Hint parallelogram: co-interior angles add up to 180°

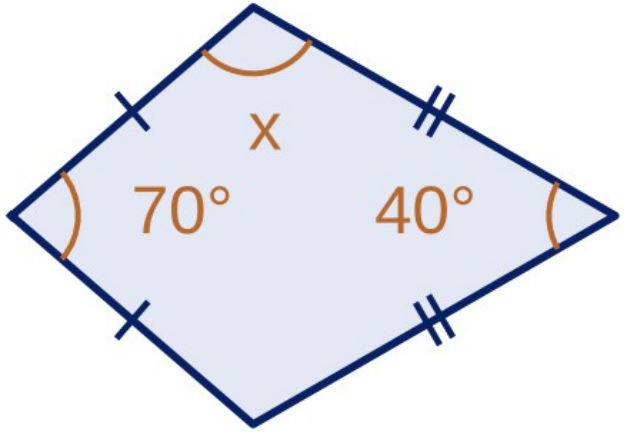
Example 1

This is a parallelogram. Find x.



Example 2

This is a kite. Find x.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Opposite angles are equal

x = 65°

Tip: opposite angles in a parallelogram are equal.

Example 2

The two side angles are equal

$2x = 360 - 70 - 40$

$= 250$

x = 125°

Tip: a kite has one pair of equal angles.

Task 1

PRACTICE · 9 MIN

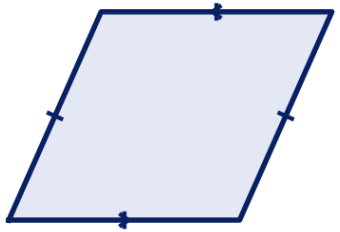
BRONZE *Answer in your book*

Hint parallelogram: co-interior angles add up to 180°

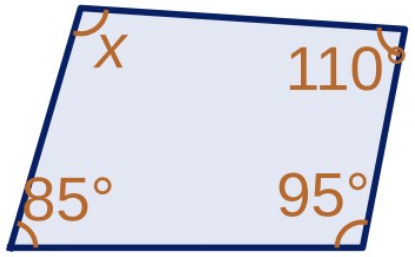
1 Which have diagonals that cross at 90°?



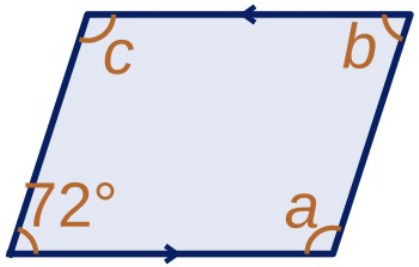
2 Name this quadrilateral.



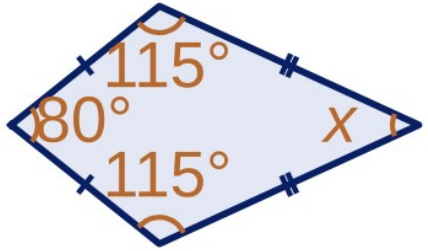
3 Find x.



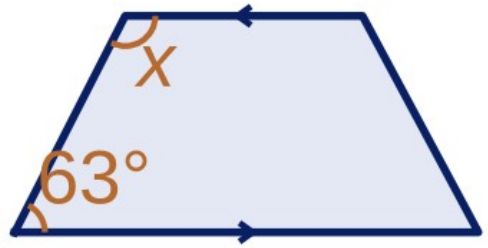
4 Find a, b and c.



5 Find x.



★ Find x.



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 **B (rhombus) and D (kite)**

2 **Rhombus**

3 $360 - 290 = \mathbf{70°}$

4 **72°, 108°, 108°**

5 $360 - 310 = \mathbf{50°}$

★ $180 - 63 = \mathbf{117°}$

Examples 3 and 4 — Using Algebra

EXAMPLES · 5 MIN

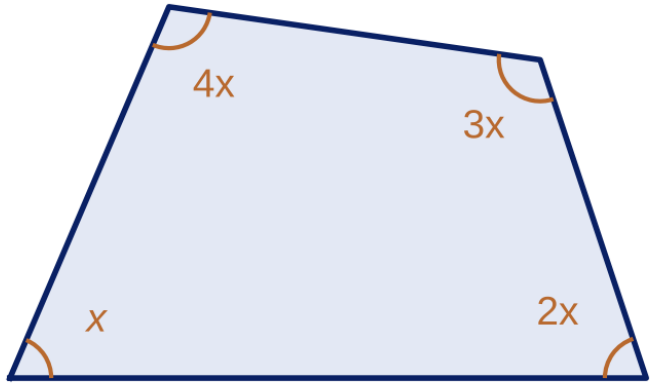
SILVER

Watch and copy into your book

Hint quadrilateral: angles add up to 360°

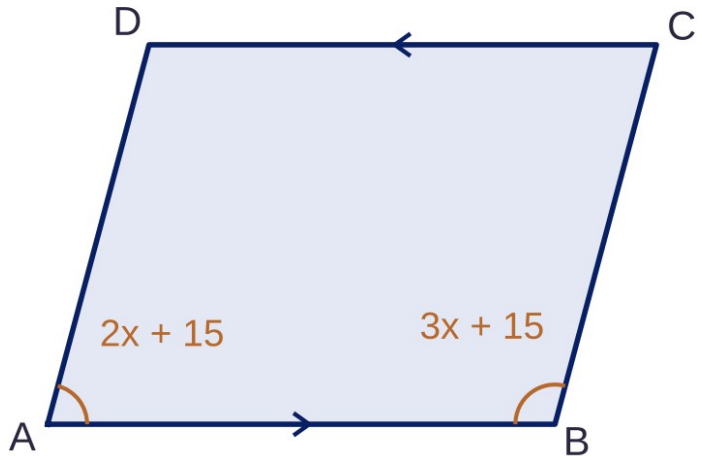
Example 3

Find x.



Example 4

ABCD is a parallelogram. Find both angles.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$x + 2x + 3x + 4x = 360$$

$$10x = 360$$

$$\mathbf{x = 36°}$$

Tip: the sum is 360°.

Example 4

$$5x + 30 = 180$$

$$x = 30$$

105° and 75°

Tip: angles next to each other add to 180°.

Task 2

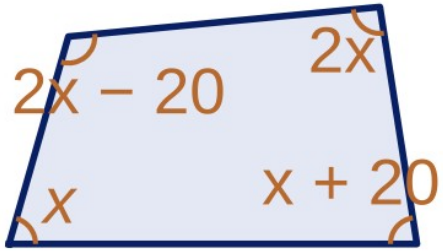
PRACTICE · 9 MIN

SILVER

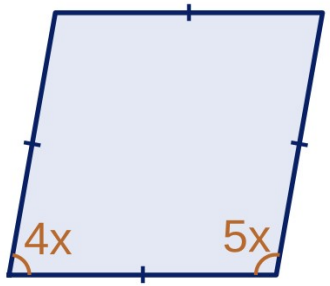
Form an equation and solve it

Hint quadrilateral: angles add up to 360°

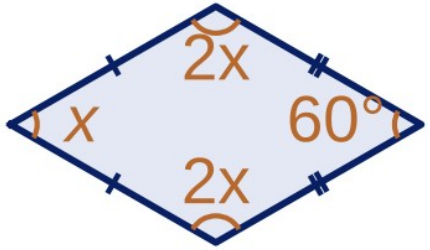
1 Find x.



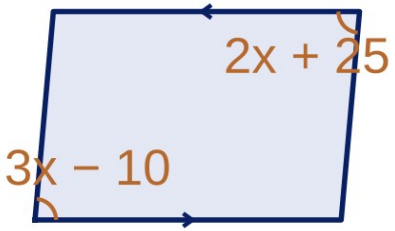
2 Rhombus. Find x.



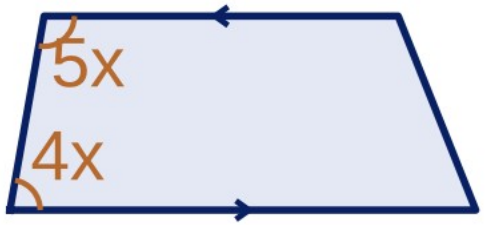
3 Kite. Find x.



4 Parallelogram. Find x.



5 Trapezium. Find x.



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $6x = 360, \quad x = \mathbf{60°}$

2. $9x = 180, \quad x = \mathbf{20°}$

3. $5x + 60 = 360, \quad x = \mathbf{60°}$

4. $3x - 10 = 2x + 25, \quad x = \mathbf{35°}$

5. $9x = 180, \quad x = \mathbf{20°}$

Examples 5 and 6 — Proofs and Problems

EXAMPLES · 5 MIN

GOLD

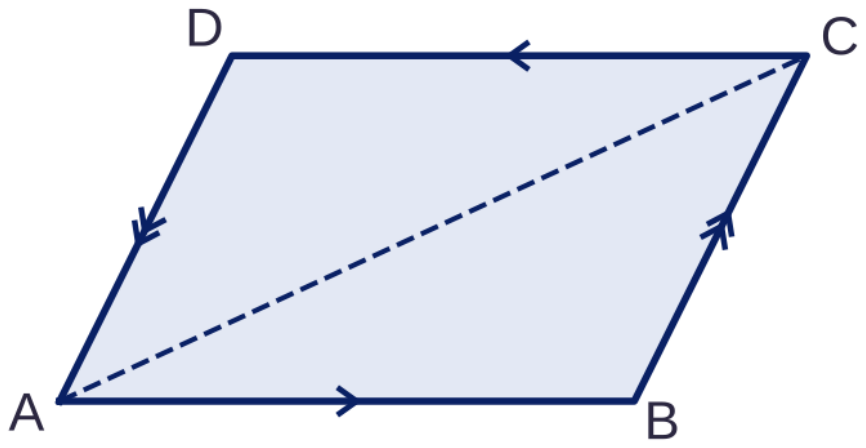
Exam-style: show all your working

Hint quadrilateral: 360° parallelogram: opposite angles equal

Example 5

(3 marks)

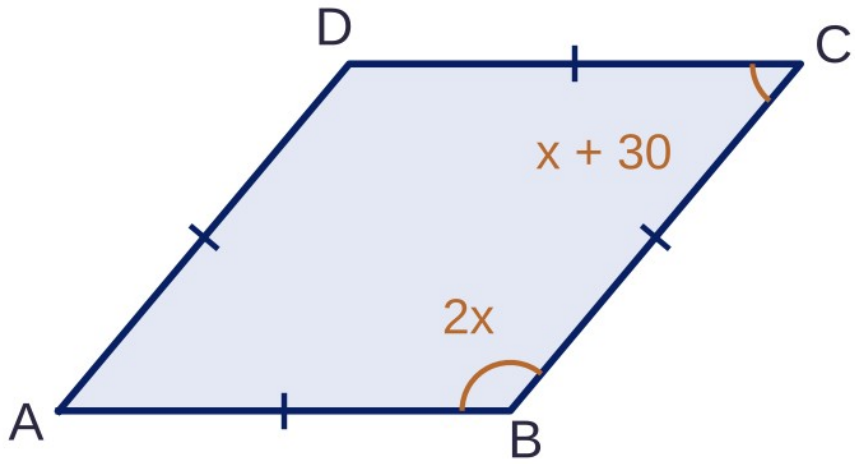
ABCD is a parallelogram.
Prove angle B = angle D.



Example 6

(3 marks)

ABCD is a rhombus. Find all the angles.



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Draw the diagonal AC

ABC ≅ CDA (ASA, alternate angles)

So angle B **= angle D**

Tip: split the shape into triangles.

Example 6

$2x + x + 30 = 180$ (co-interior)

$x = 50$

100°, 80°, 100°, 80°

Tip: angles next to each other add to 180°.

Task 3

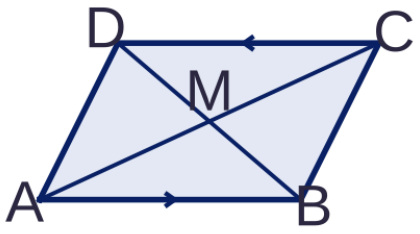
PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

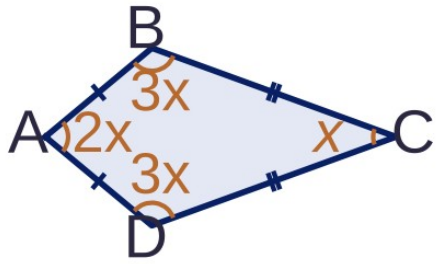
Hint quadrilateral: 360° parallelogram: opposite angles equal

1 Prove the diagonals bisect each other.



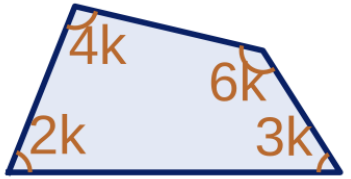
(3 marks)

2 Kite ABCD. Find x.



(3 marks)

3 Angles in the ratio 2 : 3 : 4 : 6. Find the largest.



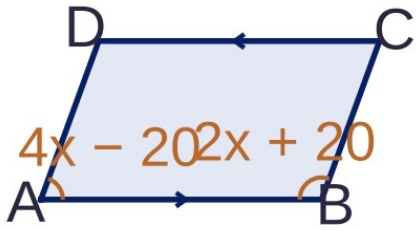
(2 marks)

4 Why is a rectangle a parallelogram?



(1 mark)

5 Parallelogram ABCD. Find angle C.



(3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. ABM ≅ CDM (ASA), AM = MC and BM = **MD**

2. $9x = 360$, $x = \mathbf{40°}$

3. $360 \div 15 = 24$, $6 \times 24 = \mathbf{144°}$

4. **Opposite sides parallel with 90° angles**

5. $6x = 180$, $x = 30$, $C = A = \mathbf{100°}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Three angles are 100°, 80° and 75°. Find the fourth.

SILVER Parallelogram: an angle is 3x and the next is 2x. Find x.

GOLD Which quadrilaterals have diagonals that bisect each other?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $360 - 255 = \mathbf{105°}$

SILVER $5x = 180, \quad x = \mathbf{36}$

GOLD **Parallelogram, rhombus, rectangle and square**

Exam Practice — Properties and Angles

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint parallelogram: co-interior angles add up to 180°

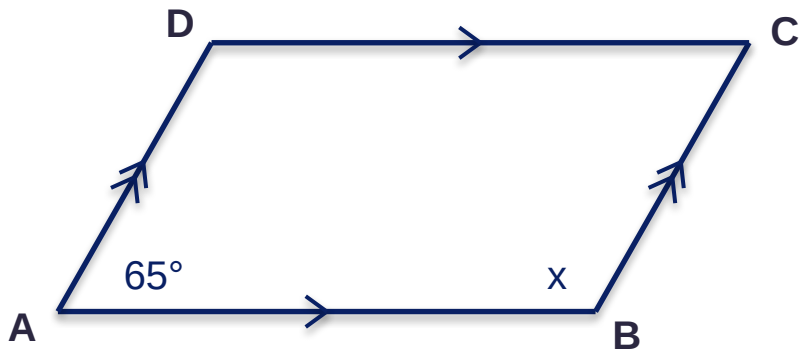
Question 1

(2 marks)

ABCD is a parallelogram.

Find x. Give a reason.

Diagram NOT accurately drawn



x = ...

Question 2

(2 marks)

Write down two properties of the diagonals of a rhombus.

.....

Exam Practice — Using Algebra

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint quadrilateral: angles add up to 360°

Question 3

(3 marks)

The angles of a quadrilateral are x°, 2x°, 100° and 80°.

Find x.

x = ...

Question 4

(3 marks)

ABCD is a kite. Angle A = (2x + 10)° and angle C = 80°.

Angles B and D are both (x + 15)°.

Find x.

x = ...

Exam Practice — Proofs and Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint quadrilateral: 360° parallelogram: opposite angles equal

Question 5

(3 marks)

The angles of a quadrilateral are in the ratio 2 : 3 : 5 : 8.

Find the largest angle.

..... °

Question 6

(3 marks)

ABCD is a parallelogram.

Angle A = (3x + 10)° and angle B = (2x + 20)°.

Find angle C.

..... °

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

x = 180 − 65 = 115° B1

co-interior angles add up to 180° B1

Tip: next-door angles add to 180°.

Question 2

they cross at 90° B1

they bisect each other B1

Tip: also: they bisect the angles.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$x + 2x + 100 + 80 = 360$ — M1

$3x = 180$ — M1

$x = 60$ — A1

Tip: angles in a quadrilateral add up to 360°.

Question 4

$2x + 10 + 80 + 2(x + 15) = 360$ — M1

$4x + 120 = 360$ — M1

$x = 60$ — A1

Tip: B and D are equal, so count (x + 15) twice.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

2 + 3 + 5 + 8 = 18 parts	M1
360 ÷ 18 = 20°	M1
8 × 20 = 160°	A1

Tip: share 360° in the ratio.

Question 6

3x + 10 + 2x + 20 = 180	M1
5x = 150, x = 30	M1
C = A = 100°	A1

Tip: opposite angles are equal.